

CML — Monitoring, Response Milestones & Resistance



1. Diagnosis baseline 100%

WHAT YOU SEE

100% marker at base of the lighthouse, DIAGNOSIS

WHAT IT ENCODES

Monitoring starts at the diagnostic baseline BCR-ABL1 ~100% on the International Scale (IS) and tracks log reductions over time by quantitative RT-PCR (the gold-standard monitoring test). Each log fall corresponds to a named molecular response (MR1=10%, MR2=1%, MR3=0.1%, MR4=0.01%, MR4.5=0.0032%).

2. 3-mo milestone $\leq 10\%$

WHAT YOU SEE

The first low rung up the lighthouse from the 100% diagnosis base — the early $\leq 10\%$ checkpoint a ship must clear at 3 months to stay on the safe course

WHAT IT ENCODES

BCR-ABL1 $\leq 10\%$ IS at 3 months is the single most important early molecular milestone — it strongly predicts long-term survival and deep response. Patients above 10% at 3 months are flagged as 'warning,' prompting adherence checks and close follow-up.

BOARD TRAP

The 3-month $\leq 10\%$ milestone is the key early predictor; $>10\%$ at 3-6 months = warning/failure. Check ADHERENCE first before assuming biologic resistance.

3. CCyR / MR2 $\leq 1\%$

WHAT YOU SEE

LEVEL 1 MR2 CCyR $\leq 1\%$ star platform

WHAT IT ENCODES

Complete cytogenetic response (CCyR) corresponds to BCR-ABL1 $\leq 1\%$ IS (MR2, a 2-log reduction) and should ideally be reached by 12 months. CCyR confers a survival benefit similar to that of the general population.

BOARD TRAP

CCyR $\approx$ BCR-ABL1 $\leq 1\%$ IS. Failure to reach $\leq 1\%$ by 12 months counts as treatment failure → mutation testing and consider switching TKI.

4. MMR / MR3 $\leq 0.1\%$

WHAT YOU SEE

LEVEL 2 MR3 MMR $\leq 0.1\%$ green platform

WHAT IT ENCODES

Major molecular response (MMR/MR3) = BCR-ABL1 $\leq 0.1\%$ IS (3-log reduction) and is the central long-term goal of TKI therapy. Achieving MMR is associated with excellent progression-free survival and a very low risk of progression.

BOARD TRAP

MMR ($\leq 0.1\%$) is the main therapeutic goal — but reaching MMR alone is NOT sufficient to stop therapy; deeper sustained response is needed for TFR.

5. No extra survival above MMR

WHAT YOU SEE

RELAPSE / NO EXTRA SURVIVAL flag at MR3 level

WHAT IT ENCODES

Once MMR is achieved, deeper responses (MR4/MR4.5) do NOT add further overall survival benefit. The value of deeper response is solely that it is the prerequisite to attempt treatment-free remission.

BOARD TRAP

Deeper than MMR (MR4 vs MR4.5) gives the SAME survival — the only reason to pursue/maintain deep molecular response is eligibility to stop therapy (TFR), not extra survival.

6. DMR / MR4-4.5 $\leq 0.01\%$

WHAT YOU SEE

LEVEL 3 MR4 DMR $\leq 0.01\%$ platform

WHAT IT ENCODES

Deep molecular response = MR4 (BCR-ABL1 $\leq 0.01\%$ IS, 4-log) or MR4.5 ($\leq 0.0032\%$, 4.5-log). Sustained DMR is the gateway to attempting treatment-free remission and reflects very low residual disease burden.

BOARD TRAP

DMR (MR4/MR4.5) is required to consider stopping a TKI; it does not by itself improve survival beyond MMR.

7. TFR (stop attempt)

WHAT YOU SEE

TFR lighthouse top, NO PILLS

WHAT IT ENCODES

Treatment-free remission (TFR) is a supervised TKI-discontinuation attempt for carefully selected patients in sustained deep molecular response. It requires very frequent PCR monitoring after stopping, with prompt TKI restart if BCR-ABL1 rises (loss of MMR).

BOARD TRAP

TFR is only for patients meeting strict criteria (typical transcript, several years of TKI, sustained DMR); it is NOT offered simply because a patient reached MMR. Most relapses occur in the first 6 months — requiring monthly PCR initially.

8. TFR criteria

WHAT YOU SEE

DOOR sign: ≥ 5 yrs TKI, DMR ≥ 2 -5 yrs, standard transcripts

WHAT IT ENCODES

Typical TFR eligibility: chronic phase only, typical/quantifiable transcript (e13a2/e14a2), ≥ 3 -5 years of TKI therapy, and a sustained deep molecular response (e.g. MR4 for ≥ 2 years), with access to reliable rapid PCR monitoring.

BOARD TRAP

All criteria must be met — atypical transcripts (can't be monitored on IS) and prior advanced phase are exclusions for a TFR attempt.

9. TFR success ~50%

WHAT YOU SEE

2 YR 50% / 5 YR 80% restart flags

WHAT IT ENCODES

About 40-50% of eligible patients maintain remission off therapy long-term; the remainder lose response (most within the first 6 months) and nearly all regain MMR after restarting the TKI. Relapse off therapy is generally not dangerous if monitored closely.

BOARD TRAP

Losing molecular response after stopping is common (~50%) but reversible — restart the TKI and the great majority regain MMR. TFR failure is not a catastrophe if caught early with monitoring.

10. 6-mo failure >10%

WHAT YOU SEE

6 MO >10% RESISTANCE rock sign

WHAT IT ENCODES

BCR-ABL1 >10% IS at 6 months indicates treatment failure/resistance and predicts poor outcome — it warrants reassessment of adherence, drug interactions, kinase-domain mutation testing, and a switch of TKI.

BOARD TRAP

Persistent >10% at 6 months = failure (not just warning). Workup order: adherence → interactions → mutation testing → switch TKI.

11. 12-mo failure >1%

WHAT YOU SEE

12 MO >1% RESISTANCE sign

WHAT IT ENCODES

Failure to achieve CCyR (BCR-ABL1 ≤1% IS) by 12 months is treatment failure. It triggers BCR-ABL1 kinase-domain mutation analysis to guide selection of the next TKI.

BOARD TRAP

Not reaching ≤1% by 12 months = failure → do mutation testing and switch; do not simply continue the same TKI at the same dose.

12. Mutation testing on failure

WHAT YOU SEE

Cluster of signposts on the rocky shore listing kinase-domain mutations, each pointing to its salvage TKI — the wreck (resistance) forces you to read the mutation map before re-routing

WHAT IT ENCODES

On resistance/failure, BCR-ABL1 kinase-domain mutation testing (Sanger or NGS) identifies the resistant clone and directs the next TKI. Different mutations have different TKI sensitivities, so the choice of salvage drug is mutation-specific.

BOARD TRAP

Always do mutation testing before switching for resistance — the next TKI is chosen by the specific mutation, not at random. Also confirm adherence, as non-adherence mimics resistance.

13. Y253H/E255K -> dasatinib

WHAT YOU SEE

Y253H, E255K -> DASATINIB sign

WHAT IT ENCODES

P-loop mutations such as Y253H and E255K confer resistance to nilotinib but remain sensitive to dasatinib — so switch to dasatinib.

BOARD TRAP

Y253H/E255K (P-loop) → choose DASATINIB, NOT nilotinib.

14. V299L/F317L -> nilotinib

WHAT YOU SEE

V299L, F317L -> NILOTINIB sign

WHAT IT ENCODES

V299L and F317L mutations confer resistance to dasatinib but remain sensitive to nilotinib — so switch to nilotinib.

BOARD TRAP

V299L/F317L (dasatinib-resistant) → choose NILOTINIB, NOT dasatinib (mirror image of the Y253H/E255K rule).

15. T315I -> ponatinib

WHAT YOU SEE

T315I -> PONATINIB sign

WHAT IT ENCODES

The T315I gatekeeper mutation is resistant to imatinib and all standard 2nd-generation TKIs. Treat with ponatinib or asciminib (or omacetaxine/HSCT), as these retain activity against T315I.

BOARD TRAP

T315I does NOT respond to imatinib, dasatinib, nilotinib, or bosutinib — only PONATINIB or ASCIMINIB (allosteric) work, with HSCT for fit candidates.